
UNIVERSITY OF CAPE TOWN

Test - 30 April 2019

Portfolio Optimization: ECO 4122Z

Time: 2 hours

Maximum Marks: 65

INSTRUCTIONS TO CANDIDATES

Answer ALL questions

1. The table below provides monthly returns for two companies listed on the JSE.

Date	ADvTECH Limited	Pan African Resources PLC
2018/05/31	-0,0095	-0,0414
2018/06/30	0,0032	0,0237
2018/07/31	-0,0291	0,0096
2018/08/31	-0,0266	0,0898
2018/09/30	-0,0278	0,0209
2018/10/31	0,0610	0,0359
2018/11/30	-0,0132	0,0140
2018/12/31	-0,0067	0,2264
2019/01/31	-0,1136	-0,0779
2019/02/28	-0,0782	-0,0888

Using the above information;

- (a) Compute the mean monthly rate of return for each stock. Which stock is most desirable by this measure? [3 marks]
- (b) Compute the standard deviation of the monthly rate of return for each stock. By this measure, which is the preferable stock? [5 marks]
- (c) Compute the coefficient of variation for each stock. By this relative measure of risk, which stock is preferable? Under what conditions must the coefficient of variation be used to measure the relative risk of two investments? [3;1 marks]

- (d) You are further told that the correlation coefficient between the monthly returns of the two stocks over the period is 0.4917. Explain the implication of this to an investor who has both stocks in their portfolio. [1 marks]

[Total: 13 marks]

2. An investor measures the utility of her wealth using the utility function $U(w) = \ln(w)$ for $w > 0$.

- (a) Derive the absolute and relative risk aversions for this investor's utility function, and the first derivative of each. [6 marks]
- (b) Comment on what this tells us about the proportion of her assets that this investor will invest in risky assets. [2 marks]

[Total: 8 marks]

3. Define the following terms:

- (a) absolute (or state-by-state) dominance [1 mark]
- (b) first order stochastic dominance [2 marks]
- (c) second order stochastic dominance [2 marks]

Consider two assets which will deliver a one-year return r_i on asset i with probabilities as set out below:

Asset	$P(r_i = -5\%)$	$P(r_i = -3\%)$	$P(r_i = 0\%)$	$P(r_i = +3\%)$	$P(r_i = +5\%)$
Asset 1	0.2	0.2	0.2	0.2	0.2
Asset 2	0.3	0.2	0.1	0.2	0.2

- (d) Compare the two assets in terms of first-order stochastic dominance [3 marks]

[Total: 8 marks]

4. Suppose that there are two risky assets: A and B , with the following properties.

	Expected Return	Standard Deviation
Asset A	0.07	0.2
Asset B	0.12	0.4

The risk-free rate is 4 percent, and assets A and B are uncorrelated.

- (a) Determine the expected return and standard deviation of a portfolio of A and B that maximizes the Sharpe ratio (i.e. optimal portfolio). [10 marks]
- (b) Investors can hold any portfolio consisting of asset A and asset B and also the risk-free asset. In addition, suppose Old Mutual asset management offers a mutual fund with an expected return of 6% and a standard deviation of 10%. All investors have standard mean-variance preferences. Should any investor invest in the Old Mutual's fund? Explain clearly why or why not. [3 marks]

[Total: 13 marks]

5. In a market where the CAPM holds there are five risky assets with the following attributes per year.

Asset number	1	2	3	4	5
Expected return	11%	10%	13%	18%	16%
Market capitalisation (in ZAR)	2.6bn	3.9bn	5.2bn		1.3bn
Beta				1.5	

The risk-free rate is $r_f = 6\%$ p.a.

- (a) Calculate the expected return on the market portfolio. [2 marks]
 (b) Deduce the market capitalisation of asset 4 and the betas of all the other assets. [6 marks]
 (c) Calculate the beta of a portfolio P which is equally weighted in the five assets and the risk-free asset. [2 marks]
 (d) State the limitations of the CAPM. [4 marks]

[Total: 14 marks]

6. In a market in which the Arbitrage Pricing Theory (APT) model holds, the expected return, $E[R_i]$, is given by

$$E[R_i] = \lambda_0 + \lambda_1 b_{i,1} + \lambda_2 b_{i,2} + \dots \lambda_n b_{i,n}$$

- (a) Define all the terms in this equation. [3 marks]

Consider a two-factor model (i.e. $n = 2$) and two well-diversified portfolios ($P1$ and $P2$) with the following features:

	P1	P2
$E[R_i]$	0.18	0.15
$b_{i,1}$	1.5	0.5
$b_{i,2}$	0.5	1.5

Let r_f denote the risk-free rate of interest and assume that $r_f = 0.075$.

- (b) Determine the values of λ_1 and λ_2 . [4 marks]

Suppose that in the market there is another portfolio with the following features:

$$E[R_3] = 0.16, b_{3,1} = 0.75, b_{3,2} = 0.7.$$

- (c) Comment on the feasibility of such a portfolio under the APT model assumptions. [3 marks]

[Total: 10 marks]